Rogue Planets & Exomoons with David Kipping

Podcast: Startalk Podcast

AI Executive Summary

1. Core Thesis

This episode synthesizes recent advancements in exoplanetary science and astrobiology through the collaboration of astrophysicist David Kipping and Neil deGrasse Tyson. The core argument focuses on redefining how we interpret exotic planetary systems (such as potential exomoons around brown dwarfs) and developing robust statistical methodologies—like cross-validation and "A-B testing"—to distinguish true biosignatures from abiotic chemical noise. Furthermore, the discussion expands the scope of habitability beyond Earth-like conditions to include rogue planets with internal heat, galactic positioning requirements for long-term survival, and the feasibility of detecting extinct civilizations through technological artifacts.

2. Key Arguments & Data

- **Exoplanet Definitions & Ambiguities:** Under IAU definitions, anything orbiting a brown dwarf (mass between ~13–80 Jupiter masses) is technically classified as a planet; however, this system (CD-35B) challenges the traditional "planet vs. star" binary by existing in a triple system configuration with high stability implications.
- **Biosignature Challenges:** Frequent false positives plague astrobiology research (e.g., Dimethyl Sulfide found on Venus-like atmospheres or K2-18b). Kipping argues for moving beyond simple gas detection toward statistical A-B testing—comparing populations with similar confounding factors but different life probabilities.
- **Super-Earth/LHS 1140b Data:** LHS 1140b has 5.5 Earth masses and 1.7 Earth radii, suggesting density that looks rocky. However, gravity compression suggests it may actually possess a gas or ice envelope rather than pure silicate rock, complicating the assumption of life support on such objects.
- **Goldilocks Zone Dynamics:** Unlike suns, M-dwarf stars have narrower and closer habitable zones due to lower luminosity; these

zones also move outward as stars age (the Sun was fainter 4 billion years ago), necessitating higher atmospheric regulation to maintain liquid water.
- **Rogue Planets & Microlensing:** It is estimated the Solar System began with ~30 planets, many ejected later. The Roman Space Telescope aims to find thousands of rogue planets using gravitational microlensing (tracking background starlight variations). These objects may remain habitable via tidal heating or internal decay rather than stellar radiation.
- **Galactic Habitable Zone:** The galaxy has a radiation environment and metallicity gradient. Civilization cannot survive too close to the Galactic Center due to supernova frequency and radiation, nor in metal-poor regions where planets cannot form. Earth's position at 2/3rds of the galactic radius is considered an ideal "suburb" for stability.
- **Technosignatures vs. Biosignatures:** Technosignatures (artificial transits, engineered satellites) are preferred over atmospheric biosignatures because they are unambiguous engineering indicators rather than chemical guesses. An archaeological contact via a time capsule might be the most probable form of discovery.
- **Funding Power Structures:** While private patrons (Patreon, Cool World donors) fuel much research, NASA and federal science agencies rely on Congressional oversight; scientific exploration requires political advocacy to prevent budget cuts at the legislative level.

3. Actionable Takeaways

- **Support Science Funding:** Contact your U.S. representatives and senators regarding the "power of the purse" to ensure continued appropriations for NASA, SETI, and major observatories like the Habitable Worlds Observatory.
- **Monitor Key Observatories:** Follow the data releases from the James Webb Space Telescope and the Roman Space Telescope to track progress in detecting rogue planets and atmospheric anomalies through cross-validation.
- **Prioritize Technosignatures:** Shift focus in astrobiological research toward identifying artificial structures or stable engineered transits

over ambiguous atmospheric gases to reduce the false-positive problem.
- **Engage with Educational Platforms:** Subscribe to educational resources like "Cool Worlds" and "Star Talk" via Patreon to support independent scientific communication and democratize access to frontier astrophysics.

Full Transcript

So Chuck, love me some David Kippen. Yes. Gets on a skateboard, comes down from Columbia, catches us up on all the cool stars out there. And I gotta tell you, you're not supposed to say it, but he's very good looking. Coming up on Star Talk. Welcome to Star Talk, your place in the universe where science and pop culture collide. Star Talk begins right now. This is star talk Neil deGrasse Tyson your personal astrophysicist got with me check nice. Yeah. What's up? What's up, Neil? Check it, baby. Hey, all right. This will be a cosmic queries. Oh, okay. Good one of our favorite people. Yeah, that's right. The boy band of astrophysicists a boy band of one Throb, no doubt We have the one, the only, David Kipping. David, welcome back to Start Talk. I don't deserve the introduction, but I will take it. Thank you. Very kind. And I have to announce that you're based in Columbia University, which is up the street a couple of miles, and you skateboarded here. There wasn't even a skateboard. What do you call the thing with the single wheel? Oh, that's a...

That's a unicycle. That's what it is. It is a unicycle. Literally. Yes, hoverboard style. So you are associate professor of astronomy up in Columbia. That's where I got my PhD at Columbia University. And you are the host of Cool World's podcast. Cool World's YouTube show and a podcast. Yes. Yes. Cool World. Same name. Cool world. Yeah, I always find people get confused. I say, even when I just had a YouTube channel and I'd say, I'd describe to people, they're like, oh, so tell me about your podcast. I'm like, no, no, that's a different, a podcast is a different thing, at least in my mind. It's a different medium. Yeah. So we have a main, why do you like these kind of mini documentaries about space and universe and things I'm working on? Then we have the podcast. So some cool world podcast and YouTube channel. You got it. There you go. So Dave, you study exoplanet. That's right. Yeah. Last I checked, we were rising through 6,000 of these things in the catalog.

pushing 7,000. I've stopped counting at this point. But I'm old enough to remember when there were none. Back in the day. Well, they were there. But we gotta get right to it. We're a planet. We're presumably an exoplanet

to aliens orbiting some other star. But we have a moon. So what's this I hear that an exo moon was recently discovered? Yeah, there was a, in the news recently, CD35, a load of letters, B, just call it CD35B for short. And CD, is this the name of the catalog it came from? That's the catalog. I've no idea what the CD is supposed to stand for, but that's the boilerplate name going around. And what's interesting about this object is it's a brown dwarf that orbits a M dwarf star. And around that brown dwarf, they see evidence for a companion. And The question is, is that a moon or is that a planet? So a brown dwarf again is a not quite a star, a little too massive to be a planet. So this is an object which is in between the mass of say 13 Jupiter masses to about 80.

And 80 is the point where you have enough mass for hydrogen fusion to happen, so you can shine like a star. But below that, you can actually still have enough mass for deuterium fusion to occur. But there's hardly any deuterium, so it's not a very efficient process. You're like a reality star. Yeah, you're on that CTL star. I was big on Bravo for a while. Right, so you know that the people who story around us get kind of... edgy about that kind of stuff, because they're like, no, no, it's not a failed star. It's an overachieving planet. So it depends how you frame it. I would rather be Jupiter. I'd rather be Jupiter. King of the planets. King of the planets instead of the overachieving, not quite a star. Do you have a red-blooded star? An almost star, not quite a star.

Too much to be a planet thing. Right. And that's not weird to have a brown dwarf orbiting a star. So this is the first time we found a basically a triple system. An object orbiting an object orbiting the object where two of them are not stars. Well, not quite because because we have binary stars. Right. No, I'm saying this is what two of them are not stars. These are two of them are not stars. Yes. So it depends on how you think of the brown dwarf. Because some might say the brown dwarf and the star is Kind of a binary star system. Yeah, but you can't, you just eliminated that possibility by degrading it to a not-quite-star overachieving planet. So did it make the news because we don't really know what to call it? It's not that it was some major astrophysical and bit of enlightenment. Is it just who ordered this kind of thing? I think it's a bit of both. I mean, first off, there is an object almost identical to this one that we discovered before.

It's like HD 206, something, something or other. That's a very similar configuration. They use a different method to find it, and they're not as confident. So they say this is a candidate object around a brown dwarf around a star. The authors of this paper are much more confident. They're like, this is legit. This is definitely something there. And so who's lead author on this? Kevin Hoy says a Chilean group using the VLT, take observations, and they're able to measure the wobble, the Doppler wobble of this object, and that's where they get this evidence for. But if you turn to the IU definitions, they're actually pretty clear, and they say that this thing's a planet. Anything in orbit for a brown dwarf is a planet according to those definitions. Now, we can argue about those definitions, but it's... Oh, so it doesn't then say, suppose the brown dwarf is orbiting something else.

doesn't say anything about it says no matter how when it says actually specifically no matter how it formed what it's doing what it's color it is whatever it is if it's between 1380 Jupiter masses it's a brown dwarf and anything in orbit of that again doesn't really matter it's always a planet wait but suppose I have a binary brown dwarf orbiting the main star you can't call that a planet no then it's a binary brown dwarf yeah because they actually do have a it's a footnote of if it's more than less than one over 25 mass ratio Then it's a binary. So this object is 1 over 37 mass ratio. So that's a moon category right there. Planet category, I'd say. There is no moon category in the IAU definition. The IAU says there's no word moon or satellite anywhere in this document. So that's why this ambiguity here. Yeah, I think so. Yeah. We should bring the Pluto haters back because they know how to get things done without a doubt.

Okay, so is it something to be excited about or is it just something to add to the catalog? I think it's definitely an exotic object. I look for Xamuns, as you know, and for me the reason why I do it is because I want to understand how common is the Earth-Moon system, how common are the moons around you, to like how common is us, like our backyard, you know, where do we come from? And this object, as cool as it is, doesn't really speak to anything familiar to that which we have in the solar system. So it's an exotic object. So that's the real, that's the rub right there. I love it. It doesn't fit the way you originally described it. It's very exotic. And that's, yeah, that's what makes it. Which is cool. I like it, but it makes it exciting. Yeah, but I also

want to understand our story. Right. Yeah. Right. How did this Chilean group discover this exo moon? I think this brown dwarf was already known and they were able to separate the light from the brown dwarf and the star using a chronograph.

and then they were able to measure the Doppler shifts of the light from that brown dwarf. Wow. So it's kind of similar to how we discovered the first exoplanet around 51 Pegasi B. That was just a star by itself and we saw the Doppler lines. Here it's harder because you've got two sources of light. So you have to do that separation first and then you can actually look for things around the brown dwarf. Wow. So but they had to know to think to do that in the first place. Yeah. Yeah. It's not easy work. So now you recently submitted for publication of paper that explores life on exoplanets, but that's quite the cottage industry. It's, you know, not a few months go by before another result. We found some chemistry in the atmosphere that means there's life on the surface, because you can't see the surface. Right. It's too small, too distant, too dim, right? But the chemistry pops out. So what are you, how are you adding to that conversation?

You know, I was just frustrated as someone interested in the search for alien life that we keep seeing so many false stars. Right? So you've had, you know, go all the way back when I was a kid, I remember Bill Clinton stood in the White House lawn talking about the Alan Hill's meteor. Yes. As evidence for life. And of course that evaporated. We've had recently K to 18B. this evidence of dimethyl sulfide that everyone got excited about could be life. And again, it's been challenged and we had venusian phosphine. So we just seem to have like time and time again. So dimethyl sulfide and it's also dimethyl disulfide, right? I think there were two. Yes. Well, the original claim is just DMS, but now they think there's that as well. So on earth, right?

phytoplankton, the boys that do total synthesis, the bases of everything here. The bases of everything, they're the ones that give us these dimethyl sulfides. And so if you're going to find that on another planet, then that's headline-making. That's really cool. So has that been retracted, that discovery? It's not been retracted, but other authors have come up with other ways you could plausibly expect to have this atmosphere without

necessarily life being involved. And you want a strong result out of here? Yeah. So this is kind of frustration we have that every, it seems like over and over again, someone sees something which looks like evidence of life and then Six months later, a theorist comes along and says, oh, by the way, I think of a way of doing that without life involved. But that doesn't make them right. It doesn't make them right. So have you found a way to make an airtight argument here? Well, I tried to point out two things. One is that as we plan these new observatories, like the Habitual Worlds Observatory, which will be the successor, hopefully, to James Webb, another multi-billion-dollar telescope we're planning right now, We have to confront the reality of this epistemic problem that will be persistent, I think, for generations to come. And I suggest that one strategy could be, kind of what we do in YouTube space, of A-B testing. So instead of just looking at a bunch of things and saying, how often do I detect dimethyl sulfide? How often do I detect a biosignature? Split your group into two categories and accept that there will be some number of confounders in these two groups, but hopefully you can design these two groups, which the confounder rate is the same, but the life rate is somehow different. If you can do that, I'm not saying it's easy to do that, but if you can do that, you can solve this problem cleanly, statistically. And that's kind of the whole premise of my paper, that we need to think about the strategy, not just the technology, but really how we even approach this question. So is that similar to... Because I remember back during the...

Was it to challenge your disaster? Any disaster, they bring in all the evidence and then come up with a conclusion. What I always think they should do is they should bring in only half the evidence and see what you conclude. And then you add to it and see if the conclusion remains. If the last bit of evidence is what tips it, then it means most of the evidence wasn't really pointing to that result. And in science, you want the more evidence to align with what your thoughts were, otherwise go home. In statistics, we call that cross-validation. Cross-validation. When you block out some of the data, you do all your fits on the data you have and then you uncover your hand and ask which one actually works. And that's like the true discriminatory test of which one is working.

Keep people on notice. I think it's the question we all care about. I think it's where there's life out there. And I just want to get an answer before I die, right? I hope that HWO succeeds. This is a major motivation of science. Yeah. I want to get it done before I die. Before I die. Right. Don't worry. When you die, Jesus will tell you all the answers. OK? We don't have to worry. I'll be right there. Space fact. If Earth rotated once in 90 minutes rather than in 24 hours. The centrifugal force of that rotation would render everyone on the equator weightless. If you like that fact, you can find 4,999 more in Lost in Space. 5,000 facts to help navigate the universe. Latest collaboration between StarTalk and National Geographic Books. Lost in Space is now available for pre-order.

wherever books are sold. This is a Cosmic Queries. You have a huge fan base out there among our followers. And so, Chuck, you got the... I... You got it lined up? I have them right here. Okay. And you haven't seen these questions, right? They're coming right out of the blue for you. Cool. Yeah, that makes three of us. And these are our Patreon supporters who have given us 5 a month. Right, which qualifies you to be able to submit your query. All right, here we go. This is Andorra Klump, who says, Dear Dr. Kipling, Dr. Tyson Lorde Nice. Kipling. Kipling, sorry. Dear Dr. Kipping, Dr. Tyson Lorde-Neiss, LHS 1140B has a confirmed atmosphere. 48 light-years away.

If microbial life lives there under ice or in its ocean, what would be the very first sign we would detect from this distance, greeting from Rotterdam, the Netherlands. And so if I remember from the news, 1140 is a rocky planet. Right. Well, we're not sure. That's a good question. We're actually not sure. Oh, really? Yeah. So it's probably almost certainly not rocky, to be honest. Really? Okay. It's about five and a half Earth masses. All right. But it's 1.7 Earth radii. Okay. So when you do the math, that gives you a density that is actually similar to that of the Earth. Sounds like it. Right. So that's why all the news was rocky planet found. Right. So you might think that's rocky, but the problem with that is is that that's That's compressed density, right? There's 1.7 Earth radii squeezing all that self-gravity. So if you kind of relaxed it, it means that the stuff that it's made of must be lighter than rock. It can't just be pure rock. It can't have squeezed in that way. Correct. So it's probably some rock and then something on top, like a big envelope of gas

or maybe an ocean or something in between like ice or something. So something's confounding those blunt calculations.

and confusing how to interpret. I mean, it's a challenge because we have nothing in the solar system to point at and say, oh, that's what it must look like. Again, that's the problem. There you go. Right, the solar system is not a thing. It's in between Earth and Neptune. And we just don't really know what those things are. Does that nullify the inquiry here? No, it's at the right distance from its star, potentially for liquid water if it has water. It's in the gold-lug zone. And it's small, so it seems like it would be potentially possible that it could have had a water delivery, it's not a gas giant. So if it has life on it, which I think is possible, it's not an earth-like planet. I want to be clear about that. It's not an earth-like planet.

it could still have life, nevertheless. And then maybe we could look for things like phytoplankton, like this dimethyl sulfide signature that we talked about earlier. We could look for oxygen, you could look for methane, you could look for some of these gases. But then it kind of comes back to that original question of like, well, then how are you sure that if you see methane, that really is life? So that's just an unsolved question. So here's what I want to know. All right, just hasn't really nothing to do with this just because you said something that triggered me. Yeah. All right. It's Goldilocks zone, right? Yeah. The star that creates the Goldilocks zone, you said the right distance. Does that distance change on the size of the star and the age of the star and do stars burn cooler when they get older, increasing or decreasing the Goldilocks zone? Yes, yes and no. Is the short answer three questions? Yeah.

There you go. That was easy. I wasn't tracking it to be that efficient in my reply. Yeah, all stars, this is actually an M dwarf. So it's a small star. I'm not sure if it's exact mass, but it's less than half the mass of the Sun. So that's why it has this red dim light. That's why it's in the LHS catalogue because it's nearby, but it's not a visible star. It's not a naked eye star. You can't see it with a naked eye. So it has to take telescopes to detect it. But it is nearby. The star, because it's less luminous than the sun, means this Goldilocks zone is much closer. Closer, right. And yeah, as it ages... And importantly, for these cooler stars, cooler, this is thousands of degrees, the Goldilocks

zone is not only closer, it's narrower. Ah, okay. Okay. Okay. So the hotter stars get bigger Goldilocks zones. Right. So these are in the statistics of Goldilocks planets, this matters. Yeah, of course, yeah. And then just to finish that point, as it ages, The Gold Oxone will actually move out usually for main sequence stars. So it should, as the Sun is increasing luminosity, it was about 30% less luminous four billion years ago than it is today. I think that's what we're talking about, right? So that means the Gold Oxone is moving out. We used to be on the outer edge of the Gold Oxone, and now the Earth is on the inner edge, which is why we're so sensitive to CO2, right? If we were two billion years ago, we could have probably polluted a lot more and not worried about it. But because we're on the inner edge of the Hubble zone now, we have to kind of worry about this a bit more. Yeah, a lot more.

No, no, yeah. Wow. We'll go back four billion years. Yeah, let's go on. Here we go. This is Alejandro Guardado. He says, he says, hello, Dr. Tyson, Lorenais and Dr. Kipping. Alejandro here from Washington state. Hello, or should I say? Hola. That is not what's in there. I know it's not. Oh, God. Okay, he says this. Let me stop messing around. I'm sorry. I gotta agree. That's what Alejandro should sound like. If he don't sound like that, he should take some lessons. I'd just see him sitting around drinking an espresso with Antonio Bandena. That's what he says. My question is, in our search for life in the universe, why does it matter if aliens are intelligent? Should the search for intelligent life be simplified to just life? Would the simplification lead to less fear in our society about aliens and lead to advances in the search? Thank you for always keeping me curious. NASA in the day used to have interest in the search for life. Okay.

which included the search for intelligent life. And then it ended up in the sights of William Proxmire for the Golden Fleece Award, which was given to agencies who do research on public monies for things that he judges is a waste of money. Golden Fleece, okay? And so NASA in the search for intelligent life, he viewed that exercise as completely beyond the pale. And so NASA in response, separated the search for life, which of interest to everyone, biologists and everybody, from the search for intelligent life. And in that separation, they forfeited the search for intelligent life, and that was picked up by SETI. Okay, so there's a whole SETI Institute, which is

privately supported. And so kind of in response to this, they are two separately funded activities. Now, if someone in your...

on your exoplanet waves to you, you don't have any problem with that. But that's not what you're after. You're looking for any kind of life at all. I think, I mean, I would especially be interested in life that was technological. I try to avoid the word intelligent because we may not even have intelligent life here. That's the answer. Yeah, that's the question. But technological life, I think, is something we could look for. And it's pretty interesting if that's out there. It does answer a different question because it's one thing to say, OK, a planet has some kind of life, microbial life. But what if in the evolutionary chain that gets from there to something like us, there is what we call a great filter, some point, which is a bottleneck, which makes it incredibly unlikely that the animals ever developed past a certain point or something. So if we discover technological life, then it would prove that, oh, there is no great filter. It's just a smooth track all the way, and it would make the so-called Fermi paradox even more puzzling. So what you're suggesting is just one other example would show that however narrow that bottleneck is it's not so narrow it would have only produced one example of technology in the universe. So if you have another example... One independent example of either life or technology would be enough to say like, hey, universe has done it twice completely independently. And life has been on earth for three and a half billion years. For two and a half of those three and a half billion, it was just single-celled organisms. So if you're throwing a dart at planets with life, And if Earth is any measure of that, you're probably going to hit a planet with microorganisms. More likely to do that. There's some of the dark statistics on this. But, okay. And would one sign of technology be smog? I think the advantage, the beauty of a techno signature is that some of them, not all of them, not smog, could be unambiguous. But if you get in like in the film Contact, where Jodie Fast has the headphones on and gets the transmission and you can unpack it and there's this engineering plan of how to build a machine, there's no way that's random chance. Someone's engineered that. Whereas if we detect a gas, so that would be slam dunk. There's definitely aliens if you detect.

dimethyl sulfide and exoplanetosphere, people are going to argue about that for years and decades to come. So that's the advantage of this thing, which

is it can be clear. They're not going to argue over engineering. Exactly. Like if they send you IKEA directions, it's a done deal. Put together your table. Right. And then at the end of the directions, it says, don't worry if there are pieces left over. Oh, wow, that's really cool. And I love that technical life. That's the way to go. All right, here we go. Larry Chan says, I'm Larry Chan from NYC and I'm a science fiction writer. Once humanity travels to other star systems, planets will most likely be named after the gods of ancient times. For example, Tyrawa and Cataclyle were Native American gods. After the list of gods reaches its limit, How would we name exoplanets? I suggest naming them after characters of classical literature. For example, Ahab, the Mad Hatter, Tom Sawyer, Alice Liddell. I love that, and there's no shortage of those names. We had a similar challenge when asteroids were discovered. How many are there? Well, there's at least 10. And they were initially named Well, the first asteroids we thought were planets. So they're named after Roman gods as are the other planets. And then we discovered there's some other category of object. But there was an urge to name them after feminized versions of people. So there's an asteroid, not called Mozart, but Mozartia.

Okay, so that was a little work, because it was the feminine would be the diminutive variant of a planet name. Okay, because they're asteroids. So then you quickly run out of names, although there's still a lot of names. I mean, there's a lot of names. People named me after their pet. There's an asteroid Santa. People love observing that on New Year's Eve. A friend of mine did that and he showed me the data. That's cool. Nerd friends, that's the kind of stuff you do. I kind of think we should just call them GPS coordinates. You give me a number and then I know exactly where to go and look at it. Because if you give me a name like go to Aurelius or something, I don't know anything about that star just from that name. I'd like to get some information about the star from the name. As you may know, the moons of Uranus and are all named for fictional characters in Shakespearean literature.

of which there are plenty enough to get you going. So another thing, they wanted to name largely successfully initially all of the asteroids whose orbit cross Earth's orbit, which puts us at risk, they're named after evil gods. That's awesome. Yes. Now that makes sense. Yes. So one of them is

Apophis, the Egyptian god of death and darkness. And there are plenty of, you know, in the polytheistic realms, there's no shortage. of these. Well, there is a shortage because there's hundreds of thousands of asteroids that'll do this. So that's the problem. But you go to literature, you've got Sauron. Sauron! That's what I'd be doing. Yeah, I love that. Well, hey, there you go. Yeah, but I think David is right. At some point, the coordinate on the sky is uniquely identifies it. Right. And you know what we can do? Because were you an amateur astronomer at all? As a kid, not.

No, at a high level. Yeah, yeah. That was a pretty high level amateur astronomer. So we, there are a lot of objects discovered by an amateur would later get a catalog designation. But then you would carry both names. Right, that was good. You say you have the coordinate as the name and then in parentheses underneath. Quotes. Tabby star, there's Barnard star. Barnard star is another one of these stars that's nearby. and Barnard first studied it, but it's got a catalog name. So I think that's how you do it. If there's a reason to have a name... because of its properties or because of who discovered it, you do it, but you still need the unambiguous identifier. Okay. Yeah. All right. All right. Hey, well, thanks a lot there, Larry. That was a good question. Okay, let's go to Mary Rose. Mary Rose says, hello to all at Start Talk. This question is coming to you from the tiny island of Malta in the Mediterranean. Ooh. I want to be there. Yes, we are. Invite us. What's your name again? Mary Rose.

Invite us all, okay? Yes, we'll be right there. Hope you got room for a sleepover. Okay, she says, David, what has been the most surprising find for you so far in the study of exoplanets? Thank you, Mary Rose. I love that. Yeah, that's a great question. There are many things we've discovered which blew our minds. I think what we kind of expected, I mean, this is a little bit for my time, is that we'd find other solar systems which look like our solar system. Right. This is the template, everything would look the same. Of course. And I think the greatest... Oh, the hubris. Yeah. We are the default. But it turns out we're actually kind of weird. And most solar systems look very different. So I think it's a very, like, kind of vague answer. But to give some examples, mini Neptune, which we talked about with this LHS 1140, that's the most common type of planet.

in the universe. What? And we do not have one. And we don't have one. We don't have one. That ain't right. That ain't right. Yeah. That's weird. That ain't right. That's kind of strange. Well, how many is a mini Neptune? It's in between about... What fraction of Neptune's mass would that be? Well, it's in between about two and four times the size of the Earth, so that's about... half the size of Neptune. Yeah, so super-Earths. Yeah. Super-Earths. Super-Earths. Is that the same as a mini-Neptune? We don't know. The other thing, because people argue about the name. Some say you can't call it super-Earth. We don't know it's rocky. You shouldn't call it mini-Neptune. I didn't want to show you to say that you don't know. You're the expert here. We don't know. That's how you know it's good. That's right. That's it. That's it. Exactly. Okay. But I think that diversity.

is, is surprising. There are certain binary planets, like the Tatooine, like your home Star Wars. Like George Lucas imagined that and everyone just thought he was crazy. But we now know that that's a very common. Do you think he really knew where did he just think it was cool? No, no, no, no. But I do. It's the only science in Star Wars. That's as good as it gets. Oh my God! It is downhill from there. That only worked because of the distance to the planet relative to the distance between the stars. So that the planet sees two stars, that's the double sunset, the famous scene, the double sunset. Those have to be so close together that the planet...

thinks they're one source of gravity. Because if they were more separated, and the planet is trying to figure out where to go to orbit, you get a chaotic trajectory. That's the three-body thing, right? It's a three-body problem. Exactly. You lose the stability and it falls into one of the planets or escapes. So that one was correctly shown with the two stars setting together, and we can judge what that distance that planet was. So then you can create stable orbit that way. So that's the only way that works. So that's good science. And just add we call that a p-type orbit. A p-type orbit. You can have a binary star where they're close together, like Tatooine that goes around the two close together on the outside. That's a p-type orbit, planet type. Or you can have the binary stars widely separated and the planet goes around around it. Oh, that's right. That's also a stable orbit. That's an s-type. That was the s-type orbit. Satellite. Oh, okay. Yeah, that's right. That's cool.

There you go. That's very cool. But then you don't get two sunsets. No, that was cool. Okay. So here's what I want to know.

I had read that one of the most common things that we never consider with planets are rogue planets, that in the formation of solar systems, so many planets are flung out of the solar system so stable orbits can be achieved by the rest of the planets, right? Yeah, they just go out of dodge. They just get out of here. So how common would it be if we were able to see them, would we just see planets wandering around like they lost from the home planets. Then they became rogue planets. Oh, well that sounds a lot more badass. It's a little bit more badass. It's only none of y'all. So David, the numbers I remember are that our solar system might have started with as many as 30 planets coming down to the eight that we now have. So that tells us that maybe there are more rogue planets then there are planets with homes. Right, yeah, this is a really hot area in astrophysics right now, rogue planets. So you could get a PhD right now doing this, it's a hot topic. But you're right, it's thought the solar system had more planets, some of them merged together. It's really thought there was a fifth giant planet. There's a lot of evidence for that. It's actually really difficult to keep Neptune stable unless there was another Neptune-like planet in the solar system in the past.

It's not canonical, but it's widely accepted that it's likely true there was a fifth giant planet in the solar system in the past that got ejected out into space. And that's how you keep the outer solar system stable. So there should be a ton of those. And the Roman space telescope, which is coming up, is going to be the perfect machine. using microlensing to detect a whole host of those. So we're expecting thousands of those to be found. Wow. So this is where you have a star in the background. Ideally, it's a dense star field like the center of the galaxy. Of course. And you just watch for any starlight that increase. No, no. Oh. That gets blighter.

Okay, oh it's brighter why because light going on either side of the planet You can't see that the rogue planet of course because it's nothing's illuminating it right in the middle of freaking nowhere Right, and there's a planet behind it pathways of that stars like going around the planet come back and join magnifying the brightness of the star itself. Because a sight line would have gone and missed you. This light line would have missed

you. They get bent into your view, adding to the brightness of the object. It's gravitational lensing for rogue planets. But it's only a tiny amount and that's what we call it micro lensing. There's enough of a variation in the travel of the light for you to determine that.

It means in terms of the magnification. It's very small but Roman's a very precise telescope. The real trick is that these are very rare events so you'd have to observe millions and millions even billions of stars simultaneously to have a chance of detecting one. per day or so, right? And that's kind of the rate they want to get. That's the power of Roman. I mean, you probably know the numbers better than I do, but the field of view is like 10 times Hubble or something like 20 times Hubble. That can boost your statistics. Well, if the star gets brighter and dimmer, how do we know it's not the star doing that? Because usually the star repeats that, whereas a microlensing event is singular.

in time. Plus, it has an exact profile. Oh, it would be singular because the planet is on the move! It's on the move! It's on the move! Oh, Christ! And the profile of the brightness and the dimming is you know that in advance because of Einstein's general relativity. So let me ask you, David. That's wild. David, Earth retains some of its heat of formation. Yeah. And it's got heat from radioactive decay within the crust. If Earth left the solar system, there would still be a source of energy within it. Some life would not care that we no longer have the sun to sustain it. How many of these rogue planets still have energy within them and therefore might still have life making no reference to the sun because life in the bottom of the ocean never seen the sun ever. Anyway, they're all blind. It's possible. I mean, people have speculated about that recently. If you have moons, it's even better.

Right, because moons can have tidal heating. Think about Io. Right. Io is a volcanic world. Lots of energy there. Europa too probably has a liquid ocean. By tidal heating you mean like a subtle stretching and contracting of the... Yeah, so as multiple moons interact with each other, it causes sometimes the moons to plunge a little bit deeper into Jupiter's gravitational field, sometimes a little bit further out. And so it's like a squash ball. It kind of gets stretched and squeezed like a piece of dough. Probably what

happened to these moons, and so you could have for billions of years life just thriving in between the stars. That is dope. Yes, totally dope. That is unbelievable, man. Well, it's totally believable. That's what makes it dope. Wow. Yeah. And in fact, look at the bias thrust upon us by biologists saying the sun is the source of all life in the world.

the source of most life. But this life, doing the backstroke at the bottom of the ocean, thriving off of geothermal energy. And so it's not that life needs the sun, it's that life needs energy. And if you find another way to get the energy, that's why we're looking for life on Europa. Another tidally heated moon of Jupiter. Interesting. Wow, that's so cool, man. So what's the difference between Europa and Io in their tidal heating? in terms of the amount of energy. Yeah. Well, Io is on the inside. It's closest to Jupiter, so it gets the real brunt of it. Yeah, you got it. Europe is, I think, number two, right? So it still gets heated, but not as much as Io. But Io is so significant that there's volcanoes on it. Yes. Yeah. Right. Yeah. It's liquefied the rocks. Oh, you can see it coming out into space. Yeah. It's crazy. Yeah. Cool. I'm Nicholas Costella, and I'm a proud supporter of Star Talk on Patreon.

This is Star Talk with Neil deGrasse Tyson. Australia, okay. I'd like to get me some dimmies. Okay. All right. He says thinking of the Fermi path. That's quite the intro to this. Did you make that up or did you write that? No, this is written. I don't know sometimes. I was just doing a terrible Australian accent. That's all I had to do that. Sometimes I don't know what it is. Okay. Okay, go on. Okay. He says, thinking of the Fermi paradox, how close can we get to SAG A? before ambient radiation from surrounding stars would make life impossible. And would pulsars, quasars and colliding black holes sterilize their galaxies. Oh, wow! So the amount of radiation coming out of a pulsar or a collision. Sagittarius A is the supermassive black hole in the center of the galaxy. And it is a rockin' sock in place. And other galaxies, it's even...

more severe than ours in terms of just the flux of high energy radiation. So do you guys think about... That's wild. Let's soften this just a little bit. Okay. Some stars give off a lot of UV, but like the blue stars, the bluer stars, and UV is hostile to life. Everything. So are you thinking about life on planets that orbit high mass, high temperature stars? Yeah. So I think, but it's a great

question. And I think it touched on the idea called the galactic capital zone. So, you can have a haplosone around a star, but there might also be a haplosone in our own galaxy. Okay. And that would be a radiation problem for you if you're a little too close to the center. Not necessarily just radiation. It could be metalicity, the metalicity of the galaxy changes. Okay. I don't know what metalicity is. It's jargon. And he's now got to explain the jargon. Okay. Okay. So, it's the heavy atomic elements. Basically, in astronomy, anything heavier than hydrogen helium, the two things came out of the Big Bang, are a metal. And that's what your...

you're made out of primarily... Chemists hate that. Because you call everything metal. Is it metal rich? Yeah. If it has, you know, carbon, it's considered a metal. And chemists hate that. So the whole universe is one big headbanger's ball. A metal man! Okay, so you have the radiation in the center, and you need enough heavy elements... To make planets? To make planets and people. And you need, I mean, when you really look around such as A-Star, you've seen some of these animations, they've reconstructed the star paths. They're on top of each other. So planets wouldn't even be stable. Yeah, okay. So all these stars... Yeah, there's like these loop orbits, and it's what won the Nobel Prize. Right. That the observations confirming the black hole in the center of the galaxy by tracking these orbits shared the Nobel Prize with our guy, our physics guy, Roger Penrose, who showed early the mathematical rigor of why you would get a black hole in the first place. The perfect combination of theory and observation there. So there's other parameters. So you've got a former planet. With the right ingredients. Yeah, you've got to have a stable orbit. And then the radiation environment has to not sterilize you. And obviously, if there's lots of stars around and say one in a thousand stars goes supernova, then the chances are if you're in a busy neighborhood, if you're in Manhattan, You'd like to have one of these guys go off at you. So you don't even want to be around another star that blows up, right? Yeah, yeah, cool. That's funny because I read that one of the reasons for life here is the fact that we're in a suburb of the Milky Way galaxy. Yeah, yeah, I think that's true. And we also don't, you know, our orbit is more or less circular. It doesn't like have an eccentric, but we plunge towards the center and come back out.

We're in a nice neighborhood. We're about two-thirds of the way out. We don't have to commute to the center. There's been red lines from where we are. We're in the Carroll Gardens Milky Way. So we're about two-thirds of the way out. And I hadn't appreciated, yes, we're in a basically circular orbit. So we don't risk... Changing neighborhoods and and surviving the consequences guys Yeah, and one full orbit is a couple hundred million years. So that's at our distance plenty of time. Well, no, but that evolutionarily, right? That's it works pretty stable. Yeah. Yeah, all right Wow, that's all cool stuff, man. Yeah. Damn. Yeah, you do some cool stuff, man I think it's called the cool laboratory. Oh, that's right. Cool laboratory. There we go. This is Mike Landers. He says, hi, Dr. Tysing, Dr. Kipping Lord NICE. This question comes from San Francisco. If we eventually detect an unambiguous technosignature or biosignature, do you think it might belong to a civilization that is already extinct by the time the signal reaches us, making our first contact essentially an archeological.

Oh, I like that. So how far away are your stars? For SETI, for technosignatures, we can search across huge spans of the galaxy. I mean, you can span tens of thousands, even 100,000 light years in some extreme cases. So they could easily have gone extinct. But your stars, your cool stars, those did not that far away. But biosignatures as things that James Webb are observing and trying to get signs of atmospheres, those are nearby. So those are... 10s, 20 light years, 30 light years away. Okay, so if presumably civilization takes so much longer than 30 years to establish that a 30 year delay is not some major... It's not prohibitive of you... Right, but the tennis signature doesn't have to be like an active beacon. It could be... One of the coolest ideas I've ever heard for a tennis signature is the idea of putting shades in space that orbit the sun and they would be artificial transits. So like put a giant triangle in space, like a sheet of Aluminium and as it transits the sun, it would create a very strange eclipse to alien observers. They'd be like, hold on, someone's put a triangle around that thing. That doesn't occur in nature. That's a space Dorido. Exactly. You'd be like, that's weird. That shouldn't be there. And that thing would be stable for millions, billions of years, long after we're gone. So we could leave a relic that would betray our presence. And you could even have interesting signals. These are the Easter Island heads.

The civilization left them there. Right. And it's been argued that you come upon the island and they're looking at you still. Right. Yeah, if you leave some purposeful relic of what you... The pyramids are a communication from another civilization to us through time. And we too have the opportunity with a time capsule of some kind to communicate to future civilizations, perhaps even future... I think the most likely concept... Wait, I need to be clear. He's not saying that pyramids were left by aliens. No. I just want to... A human civilization. Left by basically an extinct human civilization. Yes. The ancient Egypt is no longer with us. Correct. There's descendants, but they're not building pyramids anymore. My provocative claim is that the most likely alien contact we will ever have, not really a contact, but a way of them knowing that we're here, is that we could leave something like the voyage, like the golden record, like the pioneer plaques, we could leave something like that perhaps on the moon and maybe in 500 million years from now another advanced civilization will emerge and discover that on the earth. So it will be a non-human intelligence, not alien but still non-human that evolves after us and they pick this up and they realize, oh, we are not the only ones to ever develop. But that would imply that there isn't a continual awareness of a previous civilization. Are you referring to an apocalyptic earth? We all die, civilization disappears, and then some other intelligence evolves later, and discovers that... It doesn't have to be an apocalypse, but yeah. We fade out maybe, just gradually. There's evolution, you know, species very rarely last longer than a million years, so eventually we change into something else, and some other future civilization emerges. I think there's mammal species around a million.

Other species, I'm not sure. Yeah, it's rare. It's rare. There are some, but it's unusual for a species to last that long. You can't believe me. We're not going to make it. We don't even have to worry about whether or not we're going to make it until a million years. Come on. So I think that's cool. Relics are a cool way. There's a whole thing called artifact seti, which is the question asker should definitely look into. All right. Here we go. This is, here we go. This is Aidan Rodriguez who says, hello Dr. Tyson Lorde and Dr. Kipping. Greetings from Panama City, Panama, the country that gave the world... The Panama Canal. I've been a fan of you guys for many years, and I'm a Patreon supporter now. My question is, could life exist on a planet with much stronger gravity than Earth? Say two or three times stronger, if so, it

also wouldn't evolve in an atmosphere with much higher pressure than ours. Thanks so much. Keep up the amazing work. Yeah, that's pretty wild. That's a great question. I think the immediate challenge is how would you engineer a planet to have two or three times Earth gravity. Because as you increase the gravity, a planet will naturally accumulate gas around it as it's forming. So you're going to turn into a gas giant, basically. Even only three times the gravity? I think by three times, yeah. Really? I don't think there's any super-Earths that are more than like 1.7, 1.8. Also super-Earths are just slightly more muscular but not multiples. I did not know that. They didn't get that extreme. And at that point, all the fluid that has accreted, puts a pressure on the surface, and so the pressure would be super huge too. Right, and what, yeah, so the pressure would be high. I mean, it is cool. You could probably have 1.5 times Earth gravity, no problem. And what's interesting, I think, about these things. Because everyone would weigh 50% more. Unfortunately, everyone would be on a zempline. Well, I hate all these epic... Just here's the thing. Most of us already weigh 50% more. So if you want to know what it feels like to be on such a planet, there you are. Yeah, wow.

that you want to feel like it's back on Earth, lose that one-third your body weight, and then you're back to what it is on Earth. But let me tell you something cool about weight, though. Look at what's behind you, the Saturn V. Yes. Now, if you made the gravity 1.5 times heavier, that thing's not getting up. And so, if you think about the rocket equation... Which one does all the time? You think about the rockety question? Of course, yes. That thing would have to be the size of a pyramid. That's how you'd have to lose 99.999999% of your mass as fuel. So it's thought that super earth might be a prison. That civilization might be trapped there. They can never get off them because the gravity is so strong. They can't have a space station. No, they're going to get wormholes. Don't be so...

There's no natural progression the same way that we've enjoyed. Maybe there's just so much smarter... Let's do... Little Timmy in preschool just designed a new wormhole. Let's use that one instead of the other. So, no, that it isn't interesting. Given our engineering flight technologies, we would not be able to launch from an Earth that's one and a half. Look at that. You know what else I think about? If we were Venus, you would never know.

There was a night sky with stars in it right because Venus is a thick dense opaque atmosphere. There's a tarp over Venus So let me end on something that I wanted to even start with this notion of unknown unknowns. Yeah, because On Venus, like I'm saying, I know that tarp will prevent anybody from seeing anything about the universe. There'd be no astronomy. No one would have ever thought it up. Because when you look up, you just see clouds. Okay? So, to them, an unknown unknown is, because it could be a universe beyond your planet. We don't have that problem here. Is there some unknown unknown that we don't even know we don't know that puts us in a prison? That another planet freely escapes from. That is a truly frightening thought, Neil. And it reminds me of how in cosmology, cosmologists often take for granted the fact we live now, but were we to emerge 10, maybe 100 billion years in the future, when the universe has expanded so much, you would not see the galaxies at a certain point. The galaxies were expanded beyond our horizon. And so you wouldn't, you'd think the Milky Way was the universe, and you would never know.

All of that, which is out there. I think that's an amazing coincidence. In the time of Einstein's relativity, that's what anybody thought. To him, the universe was all the stars in the night sky of our Milky Way galaxy. So have you thought about a missing chapter in our book of the universe that we don't even know is not there? How can you think about that, which you do not know? It's impossible, but it's certainly possible. But we can't imagine what that might be. Wow. Okay. That we're just blithering idiots in the I was gonna say one thing is for sure. We're stupid. Thank you So David, how can we find you remind us? You can find me on the cool words YouTube channel. So it's just cool words. Yeah cool worlds It's just called cool worlds, and I've also got a pod so you can go to the cool words podcast Drive your lab. Isn't that correct? Yeah. Yeah. Yeah, we we have a donors like you have your startup patrons. We have patrons but All of that money is just used for research actually so that's kind of fun that you could you can pledge your money to go to real discovering Hopefully more and more research money is evaporate. God. Yeah, that funding model becomes more and more significant. I think so until the day you get the phone call from somewhere else a we got a lab for you here, which is probably, yeah. And you know, with that accent, he's like, see ya, bitches. No, I'm joking. So, yeah, we already know, we already have colleagues who have been cherry-

picked. Yeah, well, this is a real problem. You guys better write, you know, reach out to your Congress.

People, reach out to your senators and let them know that you want science funded in this country. And the power of the purse still resides in the Congress and not the executive branch and that you won't stand by and allow for science to be defunded, okay? So that's it. Thank you, Chuck. No worries. Are you just smart enough? So true. Funny, you should say that, Neil. That's the name of my spursal, which is right here on the StarTalk YouTube channel. Check it out. Check my Just Smart Enough. It's easy to find. It's right there. All right. So thanks, David. My pleasure. Thank you. For skateboarding. Anytime. Anytime. For one wheeling over here. For 40 blocks south. And saving the planet with your water bottle and your one wheel. That's right.

He's still exhaling CO2, though. Still work on that. This has been Star Talk at Cosmic Query's Cool Worlds Edition. Until next time, we bid you to keep looking up.